

AI Prompt-Chaining Assessment

Category: Education

Scenario: Transforming Raw Lecture Notes into a Structured Study Guide

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Scenario Overview

Goal: Take disorganized lecture notes from an introductory Cell Biology lecture and transform them into a structured, exam-ready study guide, including a self-review (critique) stage before finalizing.

Input: Raw, Messy Lecture Notes (provided to the model in Step 1)

cell theory - all living things made of cells, cells come from pre-existing cells, cell is basic unit of life. mention Schleiden Schwann Virchow contribs. prokaryotic vs eukaryotic - prok no nucleus, no membrane organelles, smaller (1-10 micrometers), examples bacteria archaea. euk has nucleus + organelles, bigger 10-100 micrometers, examples plants animals fungi protists. organelles: nucleus - control center, has DNA, nucleolus makes ribosomes. mitochondria - powerhouse, ATP via cellular respiration, has own DNA (endosymbiotic theory). ribosomes - protein synthesis, free or bound to ER. ER - rough (ribosomes, protein processing) vs smooth (lipid synthesis, detox). golgi apparatus - modifies packages ships proteins/lipids. lysosomes - digestion, break down waste (animal cells mostly). vacuoles - storage, big central vacuole in plants for turgor pressure. chloroplasts - photosynthesis, only in plants/algae, also has own DNA endosymbiotic. cell membrane - phospholipid bilayer, selective permeability, fluid mosaic model. cell wall - only plants fungi bacteria, structural support, made of cellulose in plants. cytoskeleton - microtubules microfilaments intermediate filaments, shape + movement + transport. students often confuse prokaryote vs eukaryote and mitochondria vs chloroplast function. exam will ask to compare plant vs animal cell, label diagram, explain endosymbiotic theory.

Step 1 — Structuring & Planning (Extraction)

Goal: Establish the AI's role, feed it the raw notes, and have it identify the core topics and flag likely trouble spots — before any drafting happens.

Prompt 1

Act as an expert biology tutor and instructional designer. I'm going to give you my raw, messy lecture notes from an Introduction to Cell Biology lecture. Your task: identify the 5 core topic areas a first-year student must master, and build a structured outline for a study guide that groups related concepts under clear headings, flags commonly confused concepts for special review, and notes where a comparison table would help. Keep the tone clear and exam-focused. [Insert Messy Raw Notes Here — see Input section above]

AI Output (Intermediate Response 1)

- Cell Theory & Foundational Principles — three tenets; key scientists (Schleiden, Schwann, Virchow).

- Prokaryotic vs. Eukaryotic Cells — size, nucleus/organelle presence, examples. Flag: build a comparison table — this is a top confusion point per the notes.
- Key Organelles & Their Functions — nucleus, mitochondria, ribosomes, ER (rough/smooth), Golgi, lysosomes, vacuoles, chloroplasts. Flag: mitochondria vs. chloroplast is a common mix-up — needs a callout box.
- Membranes & Structural Support — cell membrane (fluid mosaic model); cell wall (plants/fungi/bacteria only).
- Cytoskeleton & Endosymbiotic Theory — cytoskeleton roles; endosymbiotic origin of mitochondria/chloroplasts. Exam focus to build toward: plant vs. animal comparison, diagram labeling, explaining endosymbiotic theory.

Recommendation: include two comparison tables — Prokaryotic vs. Eukaryotic (Section 2) and Plant vs. Animal cell (bridging Sections 3–4) — since the notes flag both as exam targets.

Step 2 — Drafting (Generation)

Goal: Feed the Step 1 outline back to the model and generate the full study guide under strict formatting constraints.

Prompt 2

Excellent. Using only the outline you just generated, write the full study guide. Constraints: (1) use the 5 sections as headings; (2) under each heading use bullet points, each bullet no more than 2 sentences; (3) include a comparison table for Prokaryotic vs. Eukaryotic cells and a second one for Plant vs. Animal cells; (4) add a "Commonly Confused" callout box after the organelles section; (5) end with a 6-question practice quiz (mix of short-answer and compare/contrast) with an answer key; (6) do not exceed 700 words for the guide body, excluding the quiz.

AI Output (Intermediate Response 2) — Study Guide Draft

1. Cell Theory & Foundational Principles

- All living things are composed of one or more cells, and the cell is the basic unit of life.
- Cells arise only from pre-existing cells — not spontaneous generation.
- Schleiden, Schwann, and Virchow are jointly credited with establishing these tenets.

2. Prokaryotic vs. Eukaryotic Cells

- Prokaryotic cells lack a nucleus and membrane-bound organelles and are typically 1–10 μm .
- Eukaryotic cells have a true nucleus and membrane-bound organelles and are typically 10–100 μm .

Feature	Prokaryotic	Eukaryotic
Nucleus	Absent	Present
Membrane-bound organelles	Absent	Present
Typical size	1–10 μm	10–100 μm
Examples	Bacteria, Archaea	Plants, animals, fungi, protists

3. Key Organelles & Their Functions

- Nucleus: houses the cell's DNA and directs cell activity. Its nucleolus assembles ribosomes.
- Mitochondria: generate ATP through cellular respiration and carry their own DNA.
- Ribosomes: synthesize proteins and occur free in the cytoplasm or bound to the ER.
- Rough ER: studded with ribosomes and processes proteins. Smooth ER: builds lipids and detoxifies.
- Golgi apparatus: modifies, packages, and ships proteins and lipids.
- Lysosomes: break down waste and cellular debris, mainly in animal cells.
- Vacuoles: store materials. Plant cells have one large central vacuole that maintains turgor pressure.
- Chloroplasts: carry out photosynthesis in plants and algae and carry their own DNA.

Commonly Confused

Mitochondria vs. Chloroplasts: mitochondria break down sugar to release energy (respiration, in nearly all cells); chloroplasts capture light to make sugar (photosynthesis, only in plants/algae). Both are thought to have originated as free-living bacteria (endosymbiotic theory).

Prokaryote vs. Eukaryote: "pro" = before nucleus — prokaryotes evolved first and never developed one.

4. Membranes & Structural Support

- The cell membrane is a phospholipid bilayer that controls what enters and exits (fluid mosaic model).
- The cell wall provides structural support and occurs only in plants, fungi, and bacteria; in plants it is made of cellulose.

Feature	Plant Cell	Animal Cell
Cell wall	Present (cellulose)	Absent
Chloroplasts	Present	Absent
Vacuole	Large, central	Small, if present
Shape	Fixed, rectangular	Irregular, flexible

5. Cytoskeleton & Endosymbiotic Theory

- The cytoskeleton (microtubules, microfilaments, intermediate filaments) gives the cell shape and enables movement and internal transport.
- Endosymbiotic theory proposes mitochondria and chloroplasts were once free-living prokaryotes engulfed by a host cell — evidenced by their own DNA and double membranes.

Practice Quiz

- Q1. List the three tenets of cell theory.
- Q2. Name two structural differences between prokaryotic and eukaryotic cells.
- Q3. Explain why mitochondria and chloroplasts are believed to have an endosymbiotic origin.
- Q4. Compare and contrast plant and animal cells in terms of the cell wall and vacuole.
- Q5. What is the functional difference between the rough ER and the smooth ER?
- Q6. On a diagram of an animal cell, label the nucleus, mitochondria, and ribosomes, and briefly describe each one's role.

Answer Key

- A1. Cells are the basic unit of life; all living things are made of cells; cells come from pre-existing cells.
 - A2. Prokaryotes lack a nucleus and membrane-bound organelles and are smaller than eukaryotes.
 - A3. Both organelles carry their own DNA and a double membrane, suggesting they were once independent bacteria engulfed by a host cell and retained as symbionts.
 - A4. Plant cells have a cellulose cell wall and a large central vacuole for turgor pressure; animal cells lack a cell wall and have small vacuoles, if any.
 - A5. Rough ER (ribosome-studded) processes proteins; smooth ER synthesizes lipids and detoxifies.
 - A6. Nucleus = control center, houses DNA; mitochondria = ATP production; ribosomes = protein synthesis.
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Step 3 — Critique and Refinement (The Required Gate)

Goal: Force the model to evaluate its own Step 2 draft against a specific checklist and correct any misses before finalizing.

Prompt 3

Before we finalize, act as a strict academic reviewer grading this study guide against the following checklist: (1) Are all 5 outlined topics covered? (2) Are bullets kept to a maximum of 2 sentences? (3) Do both comparison tables appear, and are they accurate? (4) Does the "Commonly Confused" callout address the mitochondria/chloroplast and prokaryote/eukaryote mix-ups noted in the raw notes? (5) Are the practice-quiz questions aligned with the stated exam focus (plant vs. animal comparison, diagram labeling, endosymbiotic theory)? First, list your critique and note where the draft failed the checklist. Second, output the final, corrected version of the study guide.

AI Output (Intermediate Response 3) — Self-Critique

- Item 1 — Pass. All 5 outlined topics are present as headings in order.
- Item 2 — Partial fail. Two bullets in Section 3 (Nucleus; Rough/Smooth ER) each read as two full sentences fused with a semicolon-style clause, pushing right up against the 2-sentence limit. They should be tightened for clarity, not because they exceed the count, but because "Rough ER / Smooth ER" packs two organelles into one bullet — split for readability.
- Item 3 — Pass. Both tables are present (Section 2 and Section 4) and factually accurate.
- Item 4 — Pass. The callout explicitly covers both flagged confusions from the raw notes.
- Item 5 — Pass. Q3 tests endosymbiotic theory, Q4 tests plant vs. animal comparison, and Q6 tests diagram labeling — all three stated exam-focus areas are represented.

Correction applied: the Rough ER / Smooth ER bullet in Section 3 is split into two separate bullets in the final version below so each organelle gets its own clean, single-topic bullet.

Final, Corrected Study Guide (Section 3 excerpt — corrected)

3. Key Organelles & Their Functions (Final)

- Nucleus: houses the cell's DNA and directs cell activity. Its nucleolus assembles ribosomes.
- Mitochondria: generate ATP through cellular respiration and carry their own DNA.
- Ribosomes: synthesize proteins and occur free in the cytoplasm or bound to the ER.

- Rough ER: studded with ribosomes and processes proteins for secretion.
- Smooth ER: synthesizes lipids and handles detoxification.
- Golgi apparatus: modifies, packages, and ships proteins and lipids.
- Lysosomes: break down waste and cellular debris, mainly in animal cells.
- Vacuoles: store materials. Plant cells have one large central vacuole that maintains turgor pressure.
- Chloroplasts: carry out photosynthesis in plants and algae and carry their own DNA.

All other sections (1, 2, 4, 5, the two comparison tables, the Commonly Confused callout, and the practice quiz) passed the checklist unchanged and remain as shown in Step 2 above. That combined with this corrected Section 3 is the final submitted study guide.